CEMEX

RECEIVED
APR 23 2003
BOULDER COUNTY
LAND USE DEPT.

To: David Callahan
From: John Lohr
Subject: TDF test reports and Summary Data
Date: April 23, 2003

Attached is the information on the TDF test conducted at Lyons during November. This information should be included in the record for the Lapse Hearing in front of the Boulder County Commissioners currently scheduled for May 8, 2003.

If you have questions, please call (303-823-2109).

Sincerely,

John Lohr
Plant Manager Lyons Cement Plant
Cemex, Inc.

FLATIRONS TOXICOLOGY

Human Health Evaluation of Cement Kiln Emissions From Coal Supplemented with Natural Gas and Coal Supplemented with Tires

CEMEX, Inc. Cement Plant
Lyons, Colorado

Report prepared for:

CEMEX, Inc.
5134 Ute Highway
Lyons, Colorado 80540

Report prepared by:

Flatirons Toxicology, Inc.
2220 Howard Place
Boulder, CO 80305

J. Scott Klingensmith, Ph.D., D.A.B.T.

March 21, 2003

Executive Summary

The CEMEX portland cement plant in Lyons, CO is considering the use of scrap tires as a future supplemental fuel source. To confirm the potential impact on human health from tire-derived fuel (TDF) compared to fossil fuels alone (FF), the Lyons plant conducted a source emissions test from November 13-22, 2002. The test was conducted under normal operating conditions and measured a wide range of metals, inorganic, and organic chemicals.

- CEMEX tested for a suite of elements and compounds far more extensive than required under current regulations.
- Emissions were measured during both conditions: FF only, and FF combined with 19% TDF.
- The results of the source emission tests indicate that overall emissions are low when either FF or TDF is used and well within all applicable permit and regulatory requirements.
- Emission levels and related health risks are similar in the Lyons Plant test to those conducted at other facilities.
- The Lyons Plant results indicate that the risk to human health from the use of FF or TDF is extremely low and below regulatory concern levels.

The specific types and amounts of chemical emissions measured at the Lyons plant for both FF and TDF were generally similar to emission profiles from previous stack tests at other cement kilns around the country. However, there were also several differences in the Lyons Plant source emission worth noting. At cement kilns that have conducted comprehensive stack emission tests and human health risk assessments, the risk driver chemicals were either formaldehyde, benzene, polychlorinated biphenyls (PCBs), dioxins/furans, carcinogenic polycyclic aromatic hydrocarbons (PAHs), or a combination of these chemicals. Although some of these chemicals were detected during the Lyons Plant emissions tests, the emission rates tended to be lower than for other facilities. Furthermore, PCBs were not detected for either FF or TDF, and dioxin/furans, expressed as TCDD equivalents, were not detected during the TDF test. Conversely, the emission rate for chromium was higher for both FF and TDF at the Lyons Plant than at other facilities, although it was still low from a human health perspective.

The emission rates for these chemicals are of particular interest in terms of human health because collectively they have been shown to be responsible for 83-98% of the potential human health risks from cement kiln emissions.

The differences in emission rates between FF and TDF were relatively modest, particularly in terms of the risk driver chemicals discussed above. Formaldehyde emission rates for TDF were twice the rates for FF and chromium was about 50% higher for TDF compared to FF. Dioxin/furan (TCDD equivalents) were barely detected during the FF test and not detected in the TDF test. Carcinogenic

polycyclic aromatic hydrocarbons (PAHs expressed as benzo(a)pyrene equivalents) were detected at extremely low concentrations in both the FF and TDF tests. Benzene emission rates were higher for TDF than FF but still on the low end of the range for most cement kilns.

No additional chemicals were identified from the Lyons Plant source emission tests that are likely to be risk drivers, either on the basis of emission rate or toxicity. Since potential human health risks, including cancer, chronic and acute risks, have been shown to be negligible at several cement kilns with similar or higher emission rates than observed in the Lyons Plant source emission tests, the same general conclusion can be drawn for the use of both FF and TDF at the Lyons Plant.

The Lyons Plant emissions tests show that the TDF emission rates for the typical risk driver chemicals are in some cases higher (formaldehyde, chromium, and benzene), lower (dioxins/furans), or approximately the same (carcinogenic PAHs, and PCBs), compared to FF. However, the differences between TDF and FF are not significant from a human health perspective due to low emission rates and equivalent chemical risk profiles. Overall, the Lyons Plant emissions tests indicate no potential for adverse public health impacts from the use of TDF compared to FF.

TABLE OF CONTENTS

SECTION		PAGE
1.0	Background	1
2.0	Data Summary and Comparisons	2
3.0	Human Health Evaluation	6
4.0	Conclusions	7
5.0	References	8
6.0	Tables	9
APPENDIX A:	Report Author Background	19
APPENDIX B:	Test Sampling and Analysis (APT) Available under separate cover and by special request.	20

LIST OF TABLES

TITLE	PAGE
TABLE 1: ENERGY INPUT DATA FOR LYONS PLANT SOURCE EMISSION TESTS	10
TABLE 2: COMPARISON OF FF AND TDF EMISSION RATE DATA TO LYONS PLANT AIR PERMIT REQUIREMENTS	11
TABLE 3: COMPARISON OF FF AND TDF EMISSION RATE DATA TO NESHAP REQUIREMENTS	12
TABLE 4: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR METALS	13
TABLE 5: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR HYDROGEN HALIDES, ALDEHYDES AND KETONES, VOCS, AND PCBs	14
TABLE 6: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR DIOXINS AND FURANS (TEQ)	15
TABLE 7: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR PAHs	16
TABLE 8: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR CARCINOGENIC PAHs	17
TABLE 9: SUMMARY OF CANCER RISK DRIVERS FOR EMISSIONS FROM SEVERAL CEMENT KILNS	18

ACRONYMS

B(a)P	benzo(a)pyrene
CH3Br	bromomethane
CHCl2	methylene chloride
CS2	carbon disulfide
CO	carbon monoxide
FF	Fossil fuels (coal or natural gas)
TDF	Tire-derived fuel (coal supplemented with waste tires)
lb(s)/hr	pounds per hour
ND	not detected
ng/dscm	nanograms per dry standard cubic meter
NOX	Nitrogen oxides
PAH	polycyclic aromatic hydrocarbon
PCB	polychlorinated biphenyl
PCDD	polychlorinated dibenzodioxin
PCDF	polychlorinated dibenzofuran
SOX	Sulfur oxides
TCDD	Tetrachlorodibenzodioxin
TCDF	Tetrachlorodibenzofuran
TEQ	Toxic Equivalents Quotient
THC	Total Hydrocarbon
VOC	volatile organic chemical

1.0 Background

1.1 Cemex Lyons Cement Plant

The Lyons plant was commissioned in 1969 and has been in continuous operation for over 33 years producing portland cement. Limestone, shale, and sandstone are quarried in the Lyons area and delivered to the plant. These minerals contain the necessary proportions of calcium, silica, iron and alumina for creating the calcium silicates that is the active ingredient in portland cement. Crushed limestone, shale and sandstone are precisely metered into ball mills which pulverize it to a fine powder called raw or kiln feed. The kiln feed is introduced to the preheater/precalciner that heats the material to over 1550 degrees F, calcining it (turning it to lime), using a combination of waste heat from the kiln and additional fuel. This calcined material is then fed into the rotary kiln. The rotary kiln is a 245 foot long, 14.5 foot diameter rotating pipe. It is an effective mixer, conveyor and reaction chamber. Material is gradually heated from 1550 degrees F to over 2600 degrees F. During this process the lime turns to cement clinker with all the chemical properties of the finished product. Cement clinker is typically a black granular material with the consistency of gravel.

The clinker from the kiln is metered into the finish mill along with a small fraction of gypsum. The finish mill is a ball mill that pulverizes the clinker and gypsum into a very fine powder that is the finished product - portland cement.

A combination of fuels is used in the Lyons preheater/precalciner and kiln system. Coal and natural gas have been the most common. Tires and other materials have also been used. The fuel choice is based on a combination of the fuel's cost and chemical composition.

1.2 Tire Derived Fuel Test

Cemex expressed an interest in evaluating tires as a kiln fuel source early in 2002. This fact was widely communicated in Boulder County. Tires have been commonly used as fuel in Europe, North America and other locations since the 1970's. They carry a high heat value and are extremely volatile at high temperatures, properties important in industrial processes. Community members frequently ask questions about how the process emissions change as the fuel sources change. Extensive testing has accompanied the trial and startup of tire derived fuels at most facilities. The experiences at most facilities have been that emissions typically do not change a great deal. The concentrations of some elements and compounds go up, and some go down by small amounts. Typically, there have not been large changes in actual emissions, up or down.

Soon after announcing its interest in "tire-derived fuels" (TDF), Cemex committed that it would conduct a thorough test at the Lyons facility and publish the results prior to using TDF on an extended basis. Cemex volunteered to test for a suite of elements and compounds, far larger than required under current regulations. The intent was to conduct environmental sampling during base conditions (coal and gas only) and test conditions (tires would be added to the fuel mix). The test was planned to simulate as closely as possible the process conditions expected during extended use of TDF. The plant sustained a feed rate of tires equal to 2 tons/hour during the TDF portion of the test. This is reflective of the expected future consumption rate. An independent stack testing company collected all environmental samples. Process conditions were closely controlled by Cemex operators, engineers, and managers. State and County regulators observed or monitored stack sampling and test conditions.

1.3 Test Protocol Summary

A total of six air sampling runs were conducted over an interval of 3 days for each of the fuels sources, fossil fuel (FF) and tire-derived fuel (TDF). The following chemicals and compounds were measured near the exit of the stack; particulate matter (PM), halogen halides, aldehydes and ketones, dioxins and furans, metals, volatile organic chemicals (VOCs), polychlorinated biphenyls (PCBs) and polycyclic aromatic hydrocarbons (PAHs). Testing was conducted during normal operation. The first week of testing (November 13,15, and 22) was conducted while the kiln was fired by bituminous coal and natural gas (FF). The second week of testing (November 19, 20, and 21) was conducted while the kiln was fired by bituminous coal supplemented by tires (TDF). Details of the source emission test protocol are described in the Source Emissions Testing Report prepared by Air Pollution Testing, Inc., provided in AppendixB. Kiln operational parameters during the testing intervals are also summarized in Appendix B.

1.4 Energy Input Information

Energy inputs for the kiln during the emission tests are summarized in Table 1. For the FF emission tests, the total energy inputs averaged approximately 342 MM Btu/hr, whereas the TDF emission tests were conducted at a slightly higher average level of approximately 374 MM Btu/hr. The TDF emission tests were conducted with no natural gas and tire ratios ranging from 17.7-23.3% (MM Btu basis), with an average of 19.2% over all the runs.

2.0 Data Summaries and Comparisons

2.1 Permit requirements

2.1.1 Particulate Matter (PM)

Particulate emissions (PM) were measured using two different methods (M29 sampling train and M26A sampling train). Both of these EPA-approved methods are equivalent approaches to the measurement of PM and are frequently used together as a quality control check. As shown in Table 2, both methods were internally consistent within each test (+/- 10%) and demonstrated similar results for both FF and TDF tests. PM increased by 11% by the M29 train and decreased by 5% by the M26A train when TDF was used compared to FF. If the results of the two methods are averaged together, the PM concentrations measured during the source emissions tests represent approximately 27%-28%of the annual permit requirement for the plant for PM if extrapolated on an annual basis for FF and TDF respectively[1].

2.1.2 Criteria Pollutants

The effect of TDF on the emission rates of criteria pollutants was mixed (Table 2). Nitrogen oxides (NOx) and sulfur oxides (SOx) decreased substantially, while Total Hydrocarbons (THC) and Carbon Monoxide (CO) increased. However, all of these pollutants were measured at concentrations consistent with satisfactory compliance with the annual permit limits for the facility (personal communication from Steve Mossberg[2]).

2.1.3 VOC

VOC emission rates were higher for TDF (0.9 lb/hr) than for FF (0.2 lb/hr) but the rates were extremely low for both tests (Table 2). If these emission rates are extrapolated on an annual basis, the VOC emissions for the FF test would total 0.8 tons/yr, while TDF VOC emissions would total 3.6 tons/yr. These emissions represent 0.6% and 2.6% of the permitted VOC levels for FF and TDF respectively[1]. A more detailed comparison of VOC emission rates for the FF and TDF tests is provided in Table 5.

2.1.4 HCl

HCl emission rates were similar and very low for both the FF and the TDF test(Table 2). There are no direct emission rate limits for HCl specified in the permit[1].

2.1.5 Lead

Lead emission rates from the FF and TDF tests were low in both cases, although TDF did show an increase (Table 2). Both of these emission rates generate annual emissions for lead that are 1/1000th (FF) to 1/600th (TDF) of the current permit limits[1].

2.2 NESHAP Requirements

National emission standards for hazardous air pollutants (NESHAP)[3] are established by EPA under the Clean Air Act (1990) and are based on the best available engineering and control practices for a given industrial source. This level of control is also referred to as Maximum Achievable Control Technology or MACT.

The results from both the FF and TDF emission tests were well below applicable NESHAP requirements for PM and dioxins and furans, as shown in Table 3. When calculated in terms of pounds/ton of raw feed, PM emission rates were nearly identical (FF- 0.09; TDF - 0.10) for the two fuel tests. On a TEQ basis, dioxins and furan emission rate was 0.02 ng/m3 for the FF test, which is approximately 10 times lower than the NESHAP requirement of 0.2 ng/m^3. For the TDF emissions test, TCDD TEQ measurements were below the detection limit, which for the TDF runs was <0.02 ng/m^3.

2.3 Unregulated Air Pollutants

Relative differences in emission levels between the two tests varied by chemical class and by chemical but were very low in both cases.

2.3.1 Metals

The emission rates for metals are summarized in Table 4. Most metal emission rates increased with TDF; the increases ranged from 8-165%. Only Co (-74%)and Ag (-25%) decreased with TDF. Although the percentage differences in metal emission rates may appear rather large in some cases, it is important to note that the actual emission rates involved are extremely small. This means that the actual difference in mass of material emitted over a year's time is only a few pounds per year. For example, the maximum increase in annual mass for any one metal was a little less than 18 pounds (zinc) and the total annual increase for all metals emitted by the use of TDF was less than 100 pounds.

Typically, metal emissions contribute a small proportion (2-20 %) of the overall potential health risk based on risk assessments performed at other facilities; so, the magnitude of the increases observed with TDF are of no public health concern.

2.3.2 Hydrogen Halides

Hydrogen Chloride (HCl) was the only hydrogen halide measured in these tests. The emission rate for HCl was approximately the same (0.6 lb/hr) for both FF and TDF fuels. (Table 5).

2.3.3 Aldehydes/Ketones

Formaldehyde and acetaldehyde are commonly measured in cement kiln emissions. TDF emissions were higher than FF emissions by a factor of 2 for formaldehyde and nearly a factor of 4 for acetaldehyde (Table 5). However, the very small emission rates for these two aldehydes during the FF test exaggerate these differences. The formaldehyde emission rates

for TDF and FF at the Lyons plant are both considerably lower than those reported in most other cement kiln emission tests. Acetaldehyde is much less carcinogenic than formaldehyde and is typically not a risk driver for cement kiln emissions.

2.3.4 Volatile Organic Chemicals (VOCs)

The VOC emissions for the two fuel sources differed in both amount and types of chemicals measured. In the FF test, carbon disulfide had the highest emission rate followed by 1,4-dioxane. Small amounts of benzene, methylene chloride, ethanol and bromomethane were also detected. In contrast, in the TDF emission test the predominant VOCs were benzene and acetone with smaller amounts of chloromethane, bromomethane, toluene, tetrachlorethane, propylene, carbon disulfide, and methylene chloride. All of the VOC emission rates for both FF and TDF were relatively low from a potential health risk perspective.

2.3.5 Polychlorinated Biphenyls (PCBs)

PCBs emission rates were below detection for both FF and TDF (Table 5).

2.3.6 Dioxins/Furans

Dioxins/furans as reported in TCDD equivalents (TEQs) were low in the FF tests, (similar to previous MACT test results), and were primarily detected in Run #1 (see Appendix B). Subsequent runs with FF measured much lower levels and the results were very close to the detection limits. The average TEQ for the FF tests was 0.02 ng/m^3, which was barely above the detection limit (Table 6).

In terms of both individual congeners and total congeners, PCDD TEQs in the TDF emission tests were below the detection limit for all three runs on average.

2.3.7 Polycyclic Aromatic Hydrocarbons (PAHs)

The effect of fuel source on PAHs was mixed as shown summarized in Table 7. For example, compared to FF, TDF as the fuel source caused selected PAHs to:

- increase (naphthalene, acenapthalene, phenanthrene, chrysene).
- stay approximately the same (2 chloronapthylene, benzo(a)anthracene, benz(b)fluoranthene).
- decrease (fluorene, , benzo(g,h,i)perylene, 2-methylnapthalene).

In terms of human health risk assessment, carcinogenic PAHs (often reported in terms of benzo(a)pyrene or B(a)P equivalents[4]) are typically of most interest.

- Benzo(a)pyrene was not detected in the FF emission tests and it was detected in only one TDF run.
- While both FF and TDF emission rates for B(a)P equivalents were detected analytically, the concentrations were so low that they do not make any quantitatively significant contribution to the overall risks.

3.0 Human Health Evaluation

Based on both regulatory requirements that are intended to protect public health (permit & NESHAP) and in comparison with emission rates measured at other cement kilns, the Lyons Plant emissions are low using either FF or TDF. In fact, a comparison between emissions test data from the Lyons Plant and other cement kilns with both similar data as well as quantitative risk assessment information is a simple way to illustrate how low the Lyons Plant emissions are. One way to do this is to focus on emission rates for the small sub-set of chemicals that typically are responsible for the majority of potential health risk estimates from cement kiln emissions.

Comprehensive multi-pathway risk assessments previously conducted using source testing data from three cement kilns located in California fired with either FF or TDF have estimated potential cancer risks to range from one to nine in a million for either fuel source[5,6]. Potential non-cancer chronic and acute risks were also negligible. These risk levels are considered acceptable by EPA.

Table 9 is a brief summary of the top six risk drivers using data from three cement kiln emission tests for which comprehensive human health risk assessments have been performed. The focus of this table is hypothetical cancer risks because cancer risk estimates typically are the most conservative and generally dominate environmental human health risk assessments. This table summarizes the chemical risk drivers, emission rates, risk estimates by chemical and percentage of the total risk for these plants. The Lyons Plant emission rates have been added for comparison.

This table indicates that the risks for all four plants are driven by one or more members of a suite of six chemicals; hexavalent chromium, formaldehyde, benzene, dioxins/furans, carcinogenic PAHs, and PCBs. For the three other plants with comprehensive risk assessments, the cancer risks vary within a relatively narrow range from less than one million in a million to nine in a million for these risk driver chemicals, which is in the *de minimis*, or negligible risk range. In addition, the estimated health risks are similar for FF and TDF within a given plant. Although the pattern of the six risk drivers varies tremendously by plant and sometimes between FF and TDF, the six risk driver chemicals generally account for 80-90% or more of the overall risks.

Emission rates for these human health risk driver chemicals were generally low for the Lyons Plant, compared to the plants in California. Benzene emission rates were extremely low for TDF and barely detectable for FF. Total chromium was higher at the Lyons Plant for TDF than observed at the other plants but chromium risks were extremely low at those plants as well, so potential health risks for chromium emission rates at the Lyons Plant for either FF or TDF would still be negligible. Conversely, dioxins were lower with FF at the Lyons plant and not detected with TDF. Formaldehyde emission rates at the Lyons plant were low for both FF and TDF. Emission rates for PCBs were non-detectable for the Lyons Plant for both FF and TDF. Carcinogenic PAHs emission rates were so low that they would not make any quantitative contribution to the risk estimate.

Since the Lyons Plant emissions data for both FF and TDF are relatively low for all these risk drivers when compared to the three California cement kilns, it can be reasonably concluded that the potential health risks for the Lyons plant will also fall in the same low risk range for both these fuel sources that was calculated for those plants. This conclusion is reinforced by the identification of no additional risk driver chemicals in either the FF or TDF emission tests for the Lyons Plant that would significantly alter the risk estimate range observed at the California plants.

Specifically, the Lyons Plant emissions tests show that the TDF emission rates for the typical risk driver chemicals are in some cases higher (formaldehyde, chromium, and benzene), lower (dioxins/furans), or similar (carcinogenic PAHs and PCBs) compared to FF. However, the differences between TDF and FF are not significant from a human health perspective. This is because the emission rates for the risk driver chemicals for both FF and TDF were extremely low and the differences in emission rates were not large. Consequently, the cumulative theoretical health risk estimates for the two fuels are both negligible, approximately equivalent and for regulatory purposes, indistinguishable from each other. Overall then, the Lyons Plant emissions tests indicate no potential for adverse public health impacts from the use of TDF compared to FF.

4.0 Conclusions

The source emissions test data for the Lyons Plant indicate that the plant would remain in full compliance with all its current permit requirements if it used TDF to fire its kiln. The test data also show that FF and TDF emission rates are extremely low for all the primary human health risk driver chemicals typically found in cement kiln emissions. This is true even with the modest elevations in metal emission rates and VOCs observed with TDF. Based on all the data available, it can be concluded that there will be no significant adverse human health risks associated with the use of TDF at the Lyons Plant.

5.0 References

1. State of Colorado, Air Pollution Control Division, Colorado Operating Permit #95OPBO082, pp. 38-39

2. Personal Communication from Mr. Steve Mossberg, CEMEX Compliance Manager

3. EPA, 40 CFR Part 63, National Emission Standards for Hazardous Air Pollutants for Source Categories; Portland Cement Manufacturing Industry; Final Rule, June 1999

4. EPA, Region 9, www.epa.gov/region09/waste/sfund/prg/index.html

5. Tires as a Fuel Supplement: Feasibility Study, Report to the Legislature, January 1992, California Integrated Waste Management Board

6. AB2588 Emissions Testing at California Portland Cement Company's Colton Plant; Coal Firing and Coal with Tires Firing, Delta Air Quality Services, Inc., September 1999

TABLES

All Lyons Plant emission rate data summarized in the following tables represent the average for three analytical runs (except for FF VOCs - data from only run was available). Non-detects are replaced with one-half the detection limit for all chemicals to calculate average values except for dioxins/furans, which used the detection limit as the surrogate value of non-detect.

TABLE 1: ENERGY INPUT DATA FOR LYONS PLANT SOURCE EMISSION TESTS

Base Conditions -- Coal and Gas Only

Date	Time	Run	Tire t/hr	Tires MM Btu/hr	Kiln Coal t/hr	Calciner Coal t/hr	Calciner Nat Gas MM Btu/hr	Fossil Fuel MM Btu/hr	Tires % Total MM Btu/hr
13-Nov-02	10:07 to 16:29	Run 1	0.00	0.00	7.2	6.5	22.8	352.37	0.0%
14-Nov-02	8:24 to 12:02	Run 2	0.00	0.00	7.1	6.2	18.4	337.74	0.0%
14-Nov-02	12:53 to 16:32	Run 3	0.00	0.00	7.1	6.2	18.4	338.34	0.0%
15-Nov-02	10:36 to 12:01	Run 4	0.00	0.00	6.9	6.2	22.8	337.93	0.0%
15-Nov-02	13:15 to 14:39	Run 5	0.00	0.00	6.9	6.2	22.8	338.66	0.0%
22-Nov-02	11:52 to 13:06	Run 6	0.00	0.00	6.4	6.8	30.9	347.62	0.0%
Average for all Conditions			**0.00**	**0.00**	**6.94**	**6.34**	**22.68**	**342.11**	**0.0%**

Tire Test Conditions -- Tires and Coal

Date	Time	Run	Tire t/hr	Tires MM Btu/hr	Kiln Coal t/hr	Calciner Coal t/hr	Calciner Nat Gas MM Btu/hr	Fossil Fuel MM Btu/hr	Tires % Total MM Btu/hr
19-Nov-02	8:40 to 10:05	Run 1	2.05	65.60	7.3	5.3	0.0	304.55	17.7%
19-Nov-02	11:00 to 12:25	Run 2	2.08	66.56	7.1	5.3	0.0	297.48	18.3%
19-Nov-02	13:17 to 14:38	Run 3	2.05	65.60	7.1	5.0	0.0	290.98	18.4%
20-Nov-02	8:55 to 12:17	Run 4	2.25	72.00	6.8	6.1	0.0	311.04	18.8%
21-Nov-02	7:30 to 10:52	Run 5	2.90	92.80	6.8	5.9	0.0	305.99	23.3%
21-Nov-02	12:00 to 15:30	Run 6	2.14	68.48	6.8	5.7	0.0	300.70	18.5%
Average for all Conditions			**2.25**	**71.84**	**6.98**	**5.57**	**0.0**	**301.79**	**19.2%**

TABLE 2: COMPARISON OF FF AND TDF EMISSION RATE DATA TO LYONS PLANT AIR PERMIT REQUIREMENTS

	FF	TDF
Particulate		
particulate emissions (lb/hr) - M29 train	8.8	9.7
particulate emissions (lb/hr) - M26A train	9.5	9.1
Nox (ppm)	517.7	391.2
Sox (ppm)	26.2	15.7
THC (ppm)	0.5	1.8
CO (ppm)	50.3	120.2
Hydrogen chloride emissions (lb/hr)	0.66	0.63
Lead emissions (lb/hr)	0.0011	0.0019
VOC emissions (lb/hr)	0.2	0.9

TABLE 3: COMPARISON OF FF AND TDF EMISSION RATE DATA TO NESHAP REQUIREMENTS

	Limit	FF	TDF
Particulate Matter	0.275 lb/ton	0.09	0.10
Dioxins/Furans (TEQ)	0.2 ng/dscm	0. 02	ND*

* Dioxin/Furan TEQ Detection Limit: < 0.02 ng/dscm

TABLE 4: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR METALS

Multi-Metals (M29 Train)

	FF	TDF
antimony emissions (lb/hr)	ND	ND
arsenic emissions (lb/hr)	0.000177	0.000469
barium emissions (lb/hr)	0.00931	0.0107
beryllium emissions (lb/hr)	0.0000111	0.0000111
cadmium emissions (lb/hr)	0.000717	0.00161
chromium emissions (lb/hr)	0.000796	0.00115
cobalt emissions (lb/hr)	0.000325	0.00022
copper emissions (lb/hr)	0.000744	0.00147
lead emissions (lb/hr)	0.00111	0.00192
manganese emissions (lb/hr)	0.0073	0.00856
mercury emissions (lb/hr)	0.00669	0.00723
nickel emissions (lb/hr)	0.000882	0.00119
selenium emissions (lb/hr)	0.00177	0.00390
silver emissions (lb/hr)	0.000162	0.00012
thallium emissions (lb/hr)	0.00344	0.00535
zinc emissions (lb/hr)	0.00946	0.0117

ND - not detected; detection limits listed in Appendix 1of the Source Emissions Testing Report prepared by APT.

TABLE 5: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR HYDROGEN HALIDES, ALDEHYDES AND KETONES, VOCs, AND PCBs

	FF	TDF
Particulate		
particulate emissions (lb/hr) - M29 train	8.76	9.68
particulate emissions (lb/hr) - M26A train	9.52	9.07
Halogen Halides (M26A Train)		
hydrogen chloride emissions (lb/hr)	0.66	0.63
Aldehydes and Ketones (M0011 Train)		
formaldehyde emissions (lb/hr)	0.05	0.10
acetaldehyde emissions (lb/hr)	0.07	0.26
Speciated VOCs (Method TO-14)		
chloromethane	ND	0.005
carbon disulfide (lb/hr)	0.031	0.010
acetone (lb/hr)	ND	0.177
methylene chloride (lb/hr)	0.004	0.013
benzene (lb/hr)	0.003	0.101
toluene (lb/hr)	ND	0.008
tetrachloroethene (lb/hr)	ND	0.012
bromomethane (lb/hr)	0.005	0.006
1,4-dioxane (lb/hr)	0.026	ND
ethanol (lb/hr)	0.005	ND
propylene (lb/hr)	ND	0.007
PCBs (Method M428)	ND	ND

ND - not detected; detection limits listed in Appendix 1of the Source Emissions Testing Report prepared by APT.

TABLE 6: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR DIOXINS AND FURANS (TEQ)

PCDD/PCDF TEQs (Method M23)	**FF** (ng/mg^3)	**TDF** (ng/mg^3)
2,3,7,8-Cl4-Dibenzofuran (DB225)	ND	ND
2,3,7,8-Cl4-Dibenzo-p-dioxin	ND	ND
1,2,3,7,8-Cl5-Dibenzofuran	0.0004	ND
2,3,4,7,8-Cl5-Dibenzofuran	0.006	ND
1,2,3,7,8-Cl5-Dibenzo-p-dioxin	0.001	ND
1,2,3,4,7,8-Cl6-Dibenzofuran	0.001	ND
1,2,3,6,7,8-Cl6-Dibenzofuran	0.001	ND
2,3,4,6,7,8-Cl6-Dibenzofuran	0.001	ND
1,2,3,7,8,9-Cl6-Dibenzofuran	ND	ND
1,2,3,4,7,8-Cl6-Dibenzo-p-dioxin	0.0003	ND
1,2,3,6,7,8-Cl6-Dibenzo-p-dioxin	0.001	ND
1,2,3,7,8,9-Cl6-dibenzo-p-dioxin	0.001	ND
1,2,3,4,6,7,8-Cl7-Dibenzofuran	0.0002	ND
1,2,3,4,7,8,9-Cl7-Dibenzofuran	0.00004	ND
1,2,3,4,6,7,8-Cl7-Dibenzo-p-dioxin	0.0003	ND
1,2,3,4,6,7,8,9-Cl8-Dibenzofuran	0.00001	ND
1,2,3,4,6,7,8,9-Cl8-dibenzo-p-dioxin	0.00005	ND
Total PCDD/PCDF TEQs	0.02	ND (<0.02)

ND - not detected; detection limit used for ND in calculation of mean - detection limits listed in Appendix 1 of the Source Emissions Testing Report prepared by APT.

TABLE 7: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR PAHs

PAHs (lb/hr)	FF	TDF
napthalene	0.00045	0.0021
2-methylnapthalene	0.00035	0.00018
2-chloronapthalene	0.0000021	0.0000023
acenaphthylene	0.0000044	0.00018
acenaphthene	0.0000051	0.0000047
fluorene	0.0000061	0.0000049
phenanthrene	0.000035	0.00018
anthracene	0.0000015	0.000007
fluoranthene	0.0000061	0.000048
pyrene	0.0000075	0.000012
benz(a)anthracene	ND	ND
chrysene	0.0000039	0.000018
benzo(b)fluoranthene	0.0000012	0.0000016
benzo(k)fluoranthene	0.00000042	ND
benzo(e)pyrene	0.0000032	0.000002
benzo(a)pyrene	ND	0.000001
perylene	0.000008	ND
indeno(1,2,3-c,d)pyrene	0.0000012	ND
dibenzo(a,h)anthracene	ND	ND
benzo(g,h,i)perylene	0.000007	0.0000049

ND - not detected; detection limits listed in Appendix 1of the Source Emissions Testing Report prepared by APT.

TABLE 8: COMPARISON OF FF AND TDF EMISSION RATE DATA FOR CARCINOGENIC PAHs

PAH Results as B(a)P Equivalents (lb/hr)	B(a)P Eq	**FF**	**TDF**
benz(a)anthracene	0.1	ND	ND
chrysene	0.001	0.0000000032	0.000000012
benzo(b)fluoranthene	0.1	0.00000012	ND
benzo(k)fluoranthene	0.01	0.0000000042	0.000000016
benzo(a)pyrene	1.0	ND	0.000001
indeno(1,2,3-c,d)pyrene	0.1	0.00000012	ND
dibenzo(a,h)anthracene	1.0	ND	ND
TOTAL		0.0000002	0.000001

ND - not detected; detection limits listed in Appendix 1of the Source Emissions Testing Report prepared by APT.

TABLE 9: SUMMARY OF CANCER RISK DRIVERS FOR EMISSIONS FROM SEVERAL CEMENT KILNS

FACILITY		FF				TDF		
	CHEMICAL	*LB/HR RATE*	*RISK*	*%RISK*	*CHEMICAL*	*LB/HR RATE*	*RISK*	*%RISK*
DVNPRT	CHROMIUM	0.0000055	1.0E-08	0.50%	CHROMIUM*	0.000063	1.2E-07	13.2%
CA	FORMALDEHYDE	11.8	2.1E-06	84.0%	FORMALDEHYDE	0.51	8.9E-08	9.8%
					BENZENE	0.43	2.9E-07	31.9%
					PCB	0.00028	2.6E-07	28.6%
			2.1E-06	84.5%	SUB-TOTAL		7.6E-07	83.4%
TOTAL			2.5E-06				9.0E-07	
VCTRVLL	DIOXINS/FURANS	0.00000059	4.9E-06	79.0%	DIOXINS/FURANS	0.00000098	8.3E-06	96.5%
CA	BENZENE	0.66	7.0E-07	11.3%	BENZENE	0.15	1.6E-07	1.9%
			5.6E-06	90.3%	SUB-TOTAL		8.5E-06	98.4%
TOTAL			6.2E-06				8.6E-06	
COLTON	PAH**	0.00375	6.6E-07	46.8%	PAH**	0.001	1.9E-07	23.6%
CA	FORMALDEHYDE	0.88	2.2E-07	15.5%	FORMALDEHYDE	0.11	2.9E-08	3.60%
	BENZENE	0.265	3.2E-07	22.5%	BENZENE	0.23	2.9E-07	36.6%
	PCB	0.0003	1.1E-07	7.6%	PCB	0.00029	1.0E-07	13.1%
					DIOXINS/FURANS	0.00000008	1.2E-07	14.4%
			1.3E-06	92.4%	SUB-TOTAL		7.3E-07	91.3%
TOTAL			1.4E-06				8.0E-07	
LYONS	FORMALDEHYDE	0.05			FORMALDEHYDE	0.1		
CO	CHROMIUM*	0.00013			CHROMIUM*	0.00018		
	DIOXINS/FURANS	0.00000001			DIOXINS/FURANS	ND		
	BENZENE	0.003			BENZENE	0.101		
	PCB	ND			PCB	ND		
	PAH**	0.0000002			PAH**	0.000001		

*Ratio of Cr+6 to total Cr from DVNRPT(16%) applied to LYONS Plant

**B(a)P Equivalents

ND - not detected; detection limits listed in Appendix 1 of the Source Emissions Testing Report prepared by APT.

APPENDIX A

QUALIFICATIONS

Joseph Scott Klingensmith, Ph.D., D.A.B.T.

Dr. Klingensmith received a Ph.D. in Pharmacology and Environmental Toxicology in 1982 from the University Of Mississippi Medical Center. In 1982, he was awarded a National Institute of Environmental Health Sciences postdoctoral fellowship in environmental toxicology at Oregon State University. Dr. Klingensmith is a board-certified toxicologist with over 20 years of experience in environmental toxicology and risk assessment. He is a full member of the Society of Toxicology, American College of Toxicology, and the Society for Risk Assessment. Dr. Klingensmith has published the results of his toxicology and risk assessment research in a number of peer-reviewed journals, served on expert panels for the U.S. Navy, U.S. Army and the State of Washington, and has held principal toxicologist positions in private industry, the Washington Department of Health, and the U.S. Environmental Protection Agency. For the past 14 years, Dr. Klingensmith has been an environmental toxicology consultant based in Boulder, CO, with local, national, and international clients. He is currently President and Principal Toxicologist of his own environmental toxicology consulting firm, Flatirons Toxicology, Inc. of Boulder, CO.

Dr. Klingensmith's professional expertise is in the areas of general toxicology, risk assessment, and risk communication. He has broad experience in the derivation of human health criteria, implementation of innovative risk assessment methods for human health and ecological risks, negotiation of regulatory compliance requirements, and development of effective public outreach, medical monitoring, and risk communication programs. He has performed risk assessments in soil, food, water, air, and wildlife for hazardous waste sites, nuclear facilities, industrial plants, mining operations, landfills, incinerators, abandoned sewage treatment sites, sludge farms, chemical weapon releases and demilitarization activities, pesticide applications, manufacturing and accidental releases, as well as a number of worker, food and consumer product safety issues.

Over the past decade, Dr. Klingensmith has been actively involved in both indoor and outdoor air risk issues. He has negotiated acute and chronic human health air criteria for a number of different exposure scenarios with federal, state, and local regulatory agencies, prepared and supervised air monitoring programs, identified air pollution control strategies for remediation programs, performed risk assessments for airborne exposure to ground water contaminants, participated in a major medical monitoring program related to airborne contaminants at a major Superfund site, and has made numerous air risk assessment presentations to citizen groups.

APPENDIX B

APT Source Emission Test Report

This report is available under separate cover and by special request.